Qn. Set Code-7

Semester: 3rd
Programme: B.Tech
Branch: All Branches

AUTUMN END SEMESTER EXAMINATION-2024

3rd Semester B.Tech

PROBABILITY AND STATISTICS

MA21001 / MA 2011

(For 2024 (L.E), 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Let A and B be two mutually exclusive events such that $A \cup B = S$ and $P(A) = 3P(B)$ then find $P(B)$.

(b) Let A and B be two events with probabilities $P(A) = 0.5$, $P(B) = 0.375$ and $P(A \cap B) = 0.25$ then find $P(A' \cap B')$.

(c) If the distribution function $F(x)$ of a continuous random variable X is given as:

$$F(x) = \begin{cases} k(x-1)^4 & \text{for } 1 < x \leq 3 \\ 0 & \text{for } x \leq 1 \\ 1 & \text{for } x > 3 \end{cases}$$

Find the value of k and find the pdf $f(x)$.

(d) If the mean and variance of a random variable X are 0.5 and 3 respectively then find the mean and variance of the random variable $Y = 3X + 2$.

(e) Let X be the time between two successive arrivals at the drive-up window of a local bank. X has an exponential distribution with λ=2. Find the expected time between two successive arrivals at the drive-up window.

(f) Show that $\sigma^2 = E(X^2) - \mu^2$ where μ and σ are mean and variance of the random variable X respectively.

(g) Let the random variable X has an exponential distribution with variance 1, then find the 50th percentile of X.

(h) If the random variable X follows a Poisson distribution such that $P(X = 1) = P(X = 2)$. Find mean and $P(X = 0)$.

(i) If A, B are independent with $P(A) = 0.3$ and $P(A \cup B) = 0.8$, then find $P(B)$.

(j) Find the sample mean and sample variance for the data 2, 7, 3, 4, 5, 6

SECTION-B

2. (a) Define Poisson distribution and then find the mean and variance of Poisson distribution. [5]

(b) Define conditional probability for any two events A and B. A computer consulting firm presently has bids out on three projects. Let A_i = {awarded project i}, for i =1, 2, 3, and suppose that $P(A_1) = 0.22$, $P(A_2) = 0.25$, $P(A_3) = 0.28$, $P(A_1 \cap A_2) = 0.11$, $P(A_1 \cap A_3) = 0.05$, $P(A_2 \cap A_3) = 0.07$, $P(A_1 \cap A_2 \cap A_3) = 0.01$. Compute the probability of the following events: [5]

a) $P(A_2|A_1)$

b) $P(A_2 \cap A_3|A_1)$

c) $P(A_2 \cup A_3|A_1)$

d) $P(A_1 \cap A_2 \cap A_3|A_1 \cup A_2 \cup A_3)$

3. (a) i. Define the correlation coefficient of two random variables X and Y and then prove that [5]

$$Corrr(aX + b, cY + d) = Corr(X, Y)$$

when a and c have the same sign. What happens if if a and c have opposite sign?

ii. Find $Corr(3X, -5Y)$, when X and Y are independent.

(b) Let X is a random variable with pdf $f(x)$ defined as: [5]

$$f(x) = \begin{cases} 2kx^2 & ,0 < x < 3 \\ 0 & ,\text{otherwise} \end{cases}$$

i. Find the value of k and $E(X)$.

ii. Find the distribution function $F(X)$.

iii. Find $P(1 < X \leq 2)$ and $P(0 < X < 3)$.

4. (a) Find the maximum likelihood estimate for the parameter p of the binomial distribution. [5]

(b) For a random sample of size $n = 22$ ofa normal population with variance $\sigma^2 = 256$ and mean $\bar{x}$=64.5, construct a 95% confidence interval for the population mean μ and then find the length of the interval. [5]

[Given $P(Z > 1.96) = 0.025$)].

5. (a) The joint probability density function of the random variables X and Y is [5]

$$f(x,y) = \begin{cases} kxy & ,0 \leq x \leq 1 \quad ,0 \leq y \leq 1 \\ 0 & ,\text{otherwise.} \end{cases}$$

Find the value of k? and also, verify that $E(XY)$=$E(Y)$ $E(Y)$.

(b) Two similar Urns A and B contain 3 red and 2 white balls, 4 white and 5 red balls respectively. If a ball is selected randomly from one of the Urns, then find the probability that the Urn is B, when the ball is red. Also, the Urn is A, when the ball is white. [5]

6. (a) The joint probability distribution of the number of cars X and the number of buses Y per signal cycle at a proposed left-turn lane is displayed in the following accompanying joint probability table. [5]

$p(x,y)$		y		
		0	1	2
x	0	0.025	0.015	0.010
	1	0.050	0.030	0.020
	2	0.125	0.075	0.050
	3	0.150	0.090	0.060
	4	0.100	0.060	0.040
	5	0.050	0.030	0.020

i. What is the probability that there is exactly one car and exactly one bus during a cycle?

ii. What is the probability that there is at most one car and at most one bus during a cycle?

iii. Compute the marginal pmf of X and of Y.

iv. Are X and Y independent random variables? Explain.

(b) i. For any three events A, B, and C with $P(C) > 0$, show that [5]
$P(A \cup B|C) + P(A \cap B|C) = P(A|C) + P(B|C)$.

ii. If A and B are two independent events, then show that A' and B are also independent, where A' is complement of A.

7. (a) Obtain the line of regression of y on x and x on y for the data given below: [5]

x	1.53	1.78	2.60	2.95	3.42.
y	33.50	36.30	340.0	45.80	453.50

(b) Find the coefficient of correlation between the heights of father (X) and son (Y) from the following data: [5]

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

*****